

Universitat Politècnica de València
Master in Artificial Intelligence, Pattern Recognition and Digital Imaging
2021-2022

MACHINE TRANSLATION

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Topics

1. Introduction to machine translation
2. Statistical machine translation
3. Neural machine translation
4. Interactive machine translation
5. Advanced topics in machine translation
6. Hierarchical and syntax-based models

Scheduling

Date	Session	Teacher	Topic
9/Nov/2021	1	F. Casacuberta	Introduction to machine translation
16/Nov/2021	2	F. Casacuberta	Statistical machine translation
23/Nov/2021	3	F. Casacuberta	Neural machine translation
30/Nov/2021	4	M. Domingo	Practical exercise SMT
7/Dec/2021	5	F. Casacuberta	Neural machine translation
14/Dec/2021	6	M. Domingo	Practical exercise NMT
21/Dec/2021	7	M. Domingo	Practical exercise MMT
11/Jan/2022	8	F. Casacuberta	Interactive machine translation
18/Jan/2022	9	F. Casacuberta	Advanced topics in machine translatio
25/Jan/2022	10	J.A. Sánchez	Hierarchical and syntax-based models

Assessment

- The course must be passed through a test, courseworks and lab exercises:
 - Test (20%). February 11th, 2022.
 - Reports of 2 courseworks (60%)
 - * SMT (30%), deadline: February 11th, 2022.
 - * NMT (30%), deadline: February 11th, 2022.
 - Report of 1 labwork (20%), deadline: February 4th, 2022.
- “Actas”: February 25th, 2022.

Lab work

1. Statistical machine translation

1.1 Building a first SMT system

1.2 Experiments with Moses

<http://www.statmt.org/moses/>

2. Neural machine translation

2.1 Building a first NMT system

2.2 Experiments with NMT-Keras

https://github.com/PRHLT/nmt-keras_practicas-TA

Bibliography on MT

1. Philipp Koehn. *Statistical Machine Translation*. Cambridge University Press, 2010.
2. Philipp Koehn. *Neural Machine Translation*, Cambridge University Press. 2020.
3. S. Barrachina, O. Bender, F. Casacuberta, J. Civera, E. Cubel, S. Khadivi, A. Lagarda H. Ney, J. Tomás, and E. Vidal. *Statistical approaches to computer-assisted translation*. Computational Linguistics, 35(1):3-28, 2009.
4. Ye Jia, Ron J. Weiss, Fadi Biadisy, Wolfgang Macherey, Melvin Johnson, Zhifeng Chen, Yonghui Wu. Direct speech-to-speech translation with a sequence-to-sequence model. arXiv 2019.

Links

1. *Association for Machine Translation in the Americas (AMTA).*
URL: <http://www.amtaweb.org/>
2. *European Association for Machine Translation (EAMT).*
URL: <http://www.eamt.org/>
3. *Asia-Pacific Association for Machine Translation (AAMT).*
URL: <http://www.aamt.info/>
4. *TAUS Data Association.*
URL: <http://www.tausdata.org/>
5. *mgiza: A word alignment tool based on famous GIZA++*
URL: <https://github.com/moses-smt/mgiza>
6. *Thot: a toolkit to train phrase-based models for statistical machine translation.*
URL: <https://github.com/daormar/thot>
7. *Moses: a factored phrase-based beam-search decoder for machine translation.*
URL: <http://www.statmt.org/moses/>
8. *CasMaCat: the next generation translator's workbench.*
URL: <http://www.casmacat.eu/>
9. *NMT-Keras: Neural Machine Translation with Keras.*
URL: <https://github.com/lvapeab/nmt-keras/>
10. *MarianNMT: Fast Neural Machine Translation in C++.*
URL: <https://marian-nmt.github.io/s/>
11. *OpenNMT: An open source neural machine translation system.*
URL: <https://opennmt.net/>

Journals

- Machine Translation

<http://www.springerlink.com/content/100310/>

- Computational Linguistics

<http://www.mitpressjournals.org/loi/coli>

- Natural Language Engineering

<http://journals.cambridge.org/action/displayJournal?jid=NLE>

- Computer Speech & Language

<http://www.journals.elsevier.com/computer-speech-and-language/>

- IEEE/ACM Transactions on Audio, Speech, and Language Processing

<http://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=10376>

- Transactions of the Association for Computational Linguistics

<http://www.transacl.org/>

Main conferences

- AMTA conferences <http://www.amtaweb.org/>
- EAMT conferences <http://www.eamt.org/>
- Machine Translation Summits (MTSummit) <http://www.aamt.info/>
- Annual Meeting of the Association for Computational Linguistics (ACL) <http://www.aclweb.org/>
- Human Language Technologies: The Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT) <http://www.naacl.org/>
- European Chapter of the ACL (EACL) <http://eacl.coli.uni-saarland.de/>
- International Conference on Computational Linguistics (COLING)
- Conference on Machine Translation (Old Workshop on Statistical Machine Translation -WMT-)
- International Workshop on Spoken Language Translation (IWSLT)

- ELSNET <http://www.elsnet.org/cgi-bin/elsnet/events.pl>

- Machine Translation Archive URL: <http://www.mt-archive.info/>
- **ACL Anthology** <https://aclanthology.coli.uni-saarland.de/>

Some translation companies located in Spain

- Pangeanic <https://www.pangeanic.com/>
- Webinterpret <https://www.webinterpret.com/us/>
- Ubiquis (Celer Soluciones+Traducciones Políglota) <https://www.ubiquis.es/en/>
- Prompsit <http://www.prompsit.com/>
- Gear translations <https://www.geartranslations.com/en/>
- Bigtranslation <https://www.bigtranslation.com/>
- ...

Some demos & translators

- Neural Machine Translation

<http://casmacat.prhlt.upv.es/inmt/>

- CasMaCat: An Interactive Machine Translation Prototype

<http://casmacat.prhlt.upv.es/matecat-test/translate/demo-eutt2.xliff/es/196-tc5uxstn/demo@4002#7542>

- SisHiTra

<http://sishitra.prhlt.upv.es/>

- Moses

<http://demo.statmt.org/>

- GoogleTrans

http://www.google.com/language_tools

- Microsoft Bing

<https://www.bing.com/translator>

- DeepL

<https://www.deepl.com/en/translator>

- Translate.com

<https://www.translate.com/>